

Sequences & Patterns

ANSWER KEY



Arithmetic sequences

- 1 Find the 10th term of the sequence 5, 8, 11, 14, ...
 $a = 5, d = 3. a_{10} = 5 + 9 \times 3 = 32.$
 - 2 Find the 15th term of the sequence 7, 12, 17, 22, ...
 $a = 7, d = 5. a_{15} = 7 + 14 \times 5 = 77.$
 - 3 An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.
 $a_{20} = 4 + 19 \times 6 = 4 + 114 = 118.$
 - 4 The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.
 $3 + 4d = 23 \Rightarrow 4d = 20 \Rightarrow d = 5.$
 - 5 An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?
 $7 + (n - 1) \times 5 = 82 \Rightarrow n - 1 = 15 \Rightarrow n = 16.$ The 16th term.
 - 6 The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.
 $a + 7 \times 4 = 34 \Rightarrow a + 28 = 34 \Rightarrow a = 6.$
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Geometric sequences

- 7 Find the next term of the sequence 2, 4, 8, 16, ...
Geometric, ratio 2: $16 \times 2 = 32.$
 - 8 Find the 6th term of the sequence 3, 6, 12, 24, ...
 $a = 3, r = 2. a_6 = 3 \times 2^5 = 3 \times 32 = 96.$
 - 9 A geometric sequence has first term 5 and common ratio 3. Find the 5th term.
 $a_5 = 5 \times 3^4 = 5 \times 81 = 405.$
 - 10 The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.
 $2 \times r^3 = 54 \Rightarrow r^3 = 27 \Rightarrow r = 3.$
 - 11 A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.
 $r = 18 \div 6 = 3.$ First term $= 6 \div 3 = 2.$
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Identifying the type of sequence

- 12** Is 5, 10, 20, 40, ... arithmetic, geometric, or neither?
Each term is double the previous one ($r = 2$): geometric.
- 13** Is 6, 10, 14, 18, ... arithmetic, geometric, or neither?
Each term is 4 more than the previous one ($d = 4$): arithmetic.
- 14** Is 1, 4, 9, 16, ... arithmetic, geometric, or neither?
Differences 3, 5, 7 are not constant, and ratios are not constant either: neither (this is the sequence of perfect squares, n^2).
- 15** Is 100, 90, 81, 72.9, ... arithmetic, geometric, or neither?
Each ratio is $\frac{90}{100} = \frac{81}{90} = \frac{72.9}{81} = 0.9$: geometric with $r = 0.9$.
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Finding missing terms

- 16** Find the missing term: 3, __, 11, 15 (arithmetic).
Common difference $= (15 - 3) \div 3 = 4$. Missing term $= 3 + 4 = 7$.
- 17** Find the missing term: 4, __, 36 (geometric, all terms positive).
 $r = \sqrt{36 \div 4} = \sqrt{9} = 3$. Missing term $= 4 \times 3 = 12$.
- 18** Insert two arithmetic means between 5 and 20, forming the four-term sequence 5, __, __, 20.
Common difference $= (20 - 5) \div 3 = 5$. The sequence is 5, 10, 15, 20: the missing terms are 10 and 15.
- 19** The sequence x , $2x + 1$, $4x - 1$ is arithmetic. Find x .
Equal consecutive differences: $(2x + 1) - x = (4x - 1) - (2x + 1) \Rightarrow x + 1 = 2x - 2 \Rightarrow x = 3$.
- 20** Find the two missing terms of the geometric sequence 2, __, __, 54.
 $r^3 = 54 \div 2 = 27 \Rightarrow r = 3$. The sequence is 2, 6, 18, 54: the missing terms are 6 and 18.
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Sum of an arithmetic series

- 21** Find the sum of the first 10 terms of the sequence 2, 5, 8, 11, ...
 $a = 2$, $d = 3$, $n = 10$. $S = \frac{n}{2}(2a + (n - 1)d) = 5(4 + 27) = 5 \times 31 = 155$.
- 22** Find the sum of the first 20 positive integers.
 $S = \frac{20 \times 21}{2} = 210$.
- 23** Find the sum of the first 12 terms of the sequence 10, 7, 4, 1, ...
 $a = 10$, $d = -3$, $n = 12$. $S = 6(20 + 11 \times (-3)) = 6(20 - 33) = 6 \times (-13) = -78$.

- 24 An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.
 $S = \frac{15}{2}(2 \times 3 + 14 \times 4) = 7.5 \times (6 + 56) = 7.5 \times 62 = 465.$
- 25 Find the sum of the first 25 positive even integers (2, 4, ..., 50).
 $a = 2, d = 2, n = 25. S = \frac{25}{2}(4 + 24 \times 2) = 12.5 \times 52 = 650.$
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Special sequences

- 26 The sequence 1, 4, 9, 16, 25, ... consists of perfect squares. Find the 8th term.
The n th term is n^2 , so the 8th term is $8^2 = 64.$
- 27 Find the 6th triangular number, from the sequence 1, 3, 6, 10, 15, ...
The n th triangular number is $\frac{n(n+1)}{2}$. For $n = 6$: $\frac{6 \times 7}{2} = 21.$
- 28 A sequence starts 2, 3, 5, 8, 13, ..., where each term is the sum of the two terms before it. Find the next two terms.
Next term = $8 + 13 = 21$. Following term = $13 + 21 = 34.$
- 29 Find the 7th term of the sequence of powers of 2: 1, 2, 4, 8, 16, ...
The n th term is 2^{n-1} , so the 7th term is $2^6 = 64.$
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Visual and growing patterns

- 30 The bar chart shows the number of small squares used to build patterns 1 through 4. Predict the number of squares in pattern 6, and state the rule.
The rule is (pattern number)². Pattern 6 uses $6^2 = 36$ squares.
- 31 A pattern of dots grows as 3, 7, 11, 15, ... (each stage adds 4 dots). Find the number of dots in stage 10.
 $a = 3, d = 4. a_{10} = 3 + 9 \times 4 = 39.$
- 32 A row of matchstick triangles uses 3, 5, 7, 9, ... matchsticks for 1, 2, 3, 4 triangles. Find the number of matchsticks for 15 triangles in a row.
 $a = 3, d = 2. a_{15} = 3 + 14 \times 2 = 31.$
- 33 A row of matchstick hexagons uses 6, 11, 16, 21, ... matchsticks for 1, 2, 3, 4 hexagons. Find the matchsticks needed for 9 hexagons in a row.
 $a = 6, d = 5. a_9 = 6 + 8 \times 5 = 46.$
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Word problems

- 34** A theater has 20 seats in the first row, and each following row has 3 more seats than the row before it. How many seats are in the 12th row?
 $a = 20, d = 3. a_{12} = 20 + 11 \times 3 = 53$ seats.
- 35** A ball is dropped and after each bounce rises to half its previous height. If it first rises to 80 m after being dropped, how high does it rise after the 4th bounce?
 Heights after each bounce form a geometric sequence with $a = 80, r = \frac{1}{2}$: after bounce 4, height
 $= 80 \times \left(\frac{1}{2}\right)^4 = 80 \div 16 = 5$ m.
- 36** A savings plan deposits 100 riyals in week 1, then increases the deposit by 20 riyals each following week. Find the total saved after 8 weeks.
 $a = 100, d = 20, n = 8. S = \frac{8}{2}(200 + 7 \times 20) = 4 \times 340 = 1,360$ riyals.
- 37** A bacteria culture starts with 50 bacteria and doubles every hour. How many bacteria are there after 6 hours?
 $50 \times 2^6 = 50 \times 64 = 3,200$ bacteria.

More on series and sequences

- 38** Find the sum of the first 8 terms of the geometric sequence 1, 2, 4, 8, ...
 $a = 1, r = 2. S = \frac{a(r^8 - 1)}{r - 1} = \frac{256 - 1}{1} = 255.$
- 39** A geometric sequence has first term 4 and common ratio 2. Find the sum of its first 6 terms.
 $S = \frac{4(2^6 - 1)}{2 - 1} = 4 \times 63 = 252.$
- 40** An arithmetic sequence has 3rd term 11 and 7th term 27. Find the common difference and the first term.
 $4d = 27 - 11 = 16 \Rightarrow d = 4.$ First term $= 11 - 2 \times 4 = 3.$
- 41** The n th term of a sequence is $T_n = 3n - 2$. Find the 12th term, and state whether the sequence is arithmetic.
 $T_{12} = 3(12) - 2 = 34.$ Since T_n is linear in n , the sequence is arithmetic, with common difference 3.

Pattern reasoning

- 42** A sequence of tile patterns is built so that stage n uses $n^2 + 1$ tiles. How many tiles are needed for stage 7?
 $7^2 + 1 = 49 + 1 = 50$ tiles.
- 43** The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in an octagon ($n = 8$).
 $\frac{8 \times 5}{2} = 20$ diagonals.

- 44** A sequence follows the rule "double and add 1", starting at 3 as the 1st term. Find the 4th term.
Term 1 = 3. Term 2 = $2(3) + 1 = 7$. Term 3 = $2(7) + 1 = 15$. Term 4 = $2(15) + 1 = 31$.
- 45** The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$.
Find the number of handshakes among 10 people.
 $\frac{10 \times 9}{2} = 45$ handshakes.